Bitcoin Price Direction Classification

Assignment 2 – Machine Learning Foundations

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1. Introduction

This report presents a binary classification study on Bitcoin (BTC) daily price direction. The goal is to predict whether BTC's closing price will increase (Direction = 1) or decrease (Direction = 0) the next day, using only historical TEXTBFOHLCV (Open, High, Low, Close, Volume) data.

We engineer 26 technical features from daily BTC data spanning **February 18, 2021 to February 18, 2022** (366 trading days). After dropping rows with NaN values from rolling computations, we retain **347 clean observations**. We apply five classification methods from the course: Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), Logistic Regression, Decision Tree, and Random Forest.

The dataset is split **temporally (80/20)** to prevent look-ahead bias: the training set covers March 9 to December 10, 2021 (277 samples), and the test set covers December 11, 2021 to February 18, 2022 (70 samples). All features are standardized using statistics computed on the training set only.

2. Feature Engineering

We engineer 26 features from the raw OHLCV data. All features use only information available at time t (no look-ahead bias). The features fall into several categories: lagged returns, volume indicators, intraday measures, moving averages, momentum/trend signals, volatility measures, technical indicators (RSI, MACD, Bollinger Bands), and candlestick patterns.

Here C_t, O_t, H_t, L_t, V_t denote Close, Open, High, Low, and Volume at day t respectively.

Feature	Formula	Description
`Return_1`	$(C_t - C_{t-1})/C_{t-1} \times 100$	Previous day's return (%)
`Return_2`	Shift of `Return_1` by 1 day	Return from 2 days ago (%)
`Return_3`	Shift of `Return_1` by 2 days	Return from 3 days ago (%)
`Volume_Change`	$(V_t - V_{t-1})/V_{t-1} \times 100$	Change in trading volume (%)
`HL_Spread`	$(H_t - L_t)/C_t \times 100$	Intraday price range (%)
`OC_Return`	$(C_t - O_t)/O_t \times 100$	Open-to-close return (%)
`MA5`	Mean of last 5 closes	5-day simple moving average
`MA10`	Mean of last 10 closes	10-day simple moving average
`MA20`	Mean of last 20 closes	20-day simple moving average
`MA5_Ratio`	$(C_t/\text{MA5} - 1) \times 100$	Momentum vs. 5-day avg (%)
`MA10_Ratio`	$(C_t/\text{MA10} - 1) \times 100$	Momentum vs. 10-day avg (%)
`Momentum_5`	$(C_t - C_{t-5})/C_{t-5} \times 100$	5-day return (%)
`Gain`	max(`Return_1`, 0)	Positive return only
`Volatility_5`	Rolling std of `Return_1` (5d)	5-day return std deviation
`RSI_14`	$100 - \frac{100}{1+RS}$, $RS = \frac{\overline{\text{Gain}}}{\overline{\text{Loss}}}$	Relative Strength Index (14-day)
`MACD`	$\text{EMA}_{12} - \text{EMA}_{26}$	MACD line
`MACD_Signal`	$\text{EMA}_9(\text{MACD})$	MACD signal line
`MACD_Hist`	MACD − MACD_Signal	MACD histogram
`BB_Position`	$\frac{C - BB_{\text{lower}}}{BB_{\text{upper}} - BB_{\text{lower}}}$	Position within Bollinger Bands
`DayOfWeek`	0 (Mon) to 6 (Sun)	Day of the week
`Volume_Ratio`	$V_t/\text{MA5}(V_t)$	Volume vs. its 5-day average
`Consec_Up`	Sum of up days in last 5	Consecutive up days (5-window)
`Body`	$(C_t - O_t)/O_t$	Candlestick body
`Upper_Shadow`	$(H - \max(O, C))/C$	Upper shadow
`Lower_Shadow`	$(\min(O, C) - L)/C$	Lower shadow
`Shadow_Imbalance`	`Lower_Shadow` − `Upper_Shadow`	Net buying/selling pressure

Target variable:

$$\text{Direction} = \begin{cases} 1 & \text{if next day's close} > \text{today's close} \\ 0 & \text{otherwise} \end{cases}$$

3. Exploratory Data Analysis

3.1 Scatter Plots

We produce four scatter plots of key feature pairs, colored by Direction (blue = Up, red = Down):

- **Return_1 vs Return_2:** The two classes overlap substantially, confirming that consecutive daily returns carry weak discriminative signal. No clear linear separation is visible.
- **MA5_Ratio vs MA10_Ratio:** These momentum features are strongly correlated (points fall along a diagonal). Classes are heavily interleaved, though some concentration of Up days appears at moderate positive momentum values.
- **OC_Return vs HL_Spread:** A characteristic fan shape emerges – larger intraday spreads correspond to larger (both positive and negative) open-to-close returns. Class overlap is high.
- **Momentum_5 vs Volatility_5:** Higher volatility periods show slightly more dispersion in both classes. No obvious class separation is present.

3.2 Correlation Heatmap

The correlation matrix reveals several important patterns:

Highly correlated groups: MA5, MA10, and MA20 are near-perfectly correlated (≈ 1.0) since they are smoothed versions of the same price series. MA5_Ratio, MA10_Ratio, and Momentum_5 are also strongly correlated, capturing overlapping momentum information. Return_1 and Gain are highly correlated since $\texttt{Gain} = \max(\texttt{Return_1}, 0)$.

Low/no correlation: Return_2 and Return_3 show weak correlation with Return_1, consistent with efficient market behavior. Volume_Change has low correlation with most features, capturing a different dimension of market activity.

Volatility cluster: HL_Spread and Volatility_5 are moderately correlated – both measure turbulence via different mechanisms (intraday range vs. rolling return dispersion).

Implication: The high correlation among moving averages motivates using ratio versions (MA5_Ratio, MA10_Ratio) rather than raw values as predictors to reduce redundancy.

4. Train / Test Split

We use a **temporal split** (no shuffling) to avoid look-ahead bias.

Parameter	Value
Features used	17 (see list below)
Training samples	277 (March 9 – December 10, 2021)
Test samples	70 (December 11, 2021 – February 18, 2022)
Train class balance	50.9% up days
Test class balance	47.1% up days
Standardization	`StandardScaler` fit on training set only

Although 26 features were engineered, only **17 were selected** for the final models. Initial experiments using all 26 features led to overfitting and lower test accuracy, as the additional features (RSI_14, MACD, MACD_Signal, MACD_Hist, BB_Position, DayOfWeek, Volume_Ratio, Consec_Up, Shadow_Imbalance) introduced noise that the classifiers fit in training but that did not generalize to the test set.

The 17 features used are: `Return_1, Return_2, Return_3, Volume_Change, HL_Spread, OC_Return, MA5, MA10, MA20, MA5_Ratio, MA10_Ratio, Momentum_5, Gain, Volatility_5, Body, Upper_Shadow, Lower_Shadow`.

5. Classification Methods and Results

5.1 Linear Discriminant Analysis (LDA)

LDA assumes $\mathbf{X} \mid Y = j \sim \mathcal{N}(\boldsymbol{\mu}_j, \boldsymbol{\Sigma})$ with a common covariance matrix across classes, producing a linear decision boundary. The classifier assigns $\mathbf{x}$ to the class maximizing:

$$\mathbf{x}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_j - \tfrac{1}{2} \boldsymbol{\mu}_j^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_j + \log \pi_j$$

Metric	Value
Train Accuracy	58.48%
Test Accuracy	52.86%

	Pred Down	Pred Up
Actual Down	10	27
Actual Up	6	27

LDA achieves 52.9% test accuracy, only marginally above the 50% baseline. It shows a strong bias toward predicting 'Up' (54 out of 70 predictions), suggesting the linear boundary poorly separates the classes.

5.2 Quadratic Discriminant Analysis (QDA)

QDA allows each class its own covariance matrix $\mathbf{\Sigma}_j$, producing a quadratic (non-linear) decision boundary. A regularization parameter (`reg_param=0.1`) is used to stabilize per-class covariance estimates.

Metric	Value
Train Accuracy	60.29%
Test Accuracy	**64.29%**

	Pred Down	Pred Up
Actual Down	22	15
Actual Up	10	23

QDA is the **best performer** with 64.3% test accuracy – the only method meaningfully above the 50% baseline. It shows balanced predictions across both classes, correctly identifying 59% of down days and 70% of up days.

5.3 Logistic Regression

Logistic Regression models:

$$P(Y = 1 \mid \mathbf{X} = \mathbf{x}) = \frac{e^{\mathbf{x}^\top \boldsymbol{\beta}}}{1 + e^{\mathbf{x}^\top \boldsymbol{\beta}}}$$

and classifies based on whether this probability exceeds 0.5. Like LDA, it produces a linear decision boundary.

Metric	Value
Train Accuracy	58.48%
Test Accuracy	48.57%

	Pred Down	Pred Up
Actual Down	9	28
Actual Up	8	25

Logistic Regression achieves 48.6% test accuracy, falling *below* the 50% baseline. Like LDA, it heavily favors predicting 'Up'. The identical training accuracy to LDA (58.5%) is expected since both methods produce linear boundaries.

5.4 Decision Tree

A Decision Tree classifier with `max_depth=5` provides a non-linear, interpretable model. The tree recursively partitions the feature space using axis-aligned splits.

Metric	Value
Train Accuracy	75.09%
Test Accuracy	51.43%

	Pred Down	Pred Up
Actual Down	24	13
Actual Up	21	12

The Decision Tree shows clear overfitting: 75.1% training accuracy drops to 51.4% on the test set (overfit gap of **23.7 pp**).

5.5 Random Forest

A Random Forest with 200 trees and `max_depth=5` aggregates many decorrelated decision trees to reduce variance.

Metric	Value
Train Accuracy	94.95%
Test Accuracy	55.71%

	Pred Down	Pred Up
Actual Down	20	17
Actual Up	14	19

Random Forest exhibits the most severe overfitting: 94.95% training accuracy plummets to 55.7% on the test set (overfit gap of **39.2 pp**). Despite the ensemble approach, the model memorizes training noise rather than capturing generalizable patterns.

6. Comparison of All Methods

Method	Train Acc.	Test Acc.	Overfit Gap
QDA	60.3%	**64.3%**	−4.0 pp
Random Forest	94.9%	55.7%	+39.2 pp
LDA	58.5%	52.9%	+5.6 pp
Decision Tree	75.1%	51.4%	+23.7 pp
Logistic Regression	58.5%	48.6%	+9.9 pp

7. Discussion

1. QDA is the clear best performer. QDA achieves 64.3% test accuracy – the only method meaningfully above the 50% random-guessing baseline. Notably, QDA exhibits a *negative* overfit gap (−4.0%), meaning it generalizes better than its training performance suggests. This indicates that the two classes have genuinely different covariance structures in feature space, and the quadratic decision boundary captures this heterogeneity more faithfully than a linear one.

2. LDA and Logistic Regression produce similar results. Both achieve identical training accuracy (58.5%), which is expected since both produce linear decision boundaries (from generative vs. discriminative assumptions). On the test set, LDA slightly outperforms Logistic Regression (52.9% vs. 48.6%). Logistic Regression falls below 50%, suggesting the linear boundary learned on training data does not transfer well to the test period.

3. Random Forest overfits most severely. With a 39.2 percentage-point gap between training (94.9%) and test (55.7%) accuracy, Random Forest memorizes the training set almost perfectly but captures little generalizable signal. This is a textbook case of overfitting on noisy, weakly-patterned data.

4. Decision Tree also overfits substantially. The Decision Tree shows a 23.7-point overfit gap (75.1% train vs. 51.4% test), confirming that tree-based methods are susceptible to fitting noise in low signal-to-noise financial data, even with depth constraints.

5. Overfit gap as a diagnostic. The three methods with the smallest gaps – QDA (−4.0%), LDA (+5.6%), and Logistic Regression (+9.9%) – are parametric models with low capacity, while tree-based methods have higher capacity and correspondingly larger gaps. This illustrates the **bias-variance tradeoff**: more flexible models fit training data better but generalize worse when the true signal is weak.

6. Difficulty of financial prediction. Achieving >60% accuracy on BTC daily direction prediction is inherently challenging because financial markets are highly efficient – past price information is quickly absorbed. That QDA reaches 64.3% is a moderately encouraging result, possibly reflecting genuine non-linear structure in the class-conditional distributions that simpler linear methods miss.